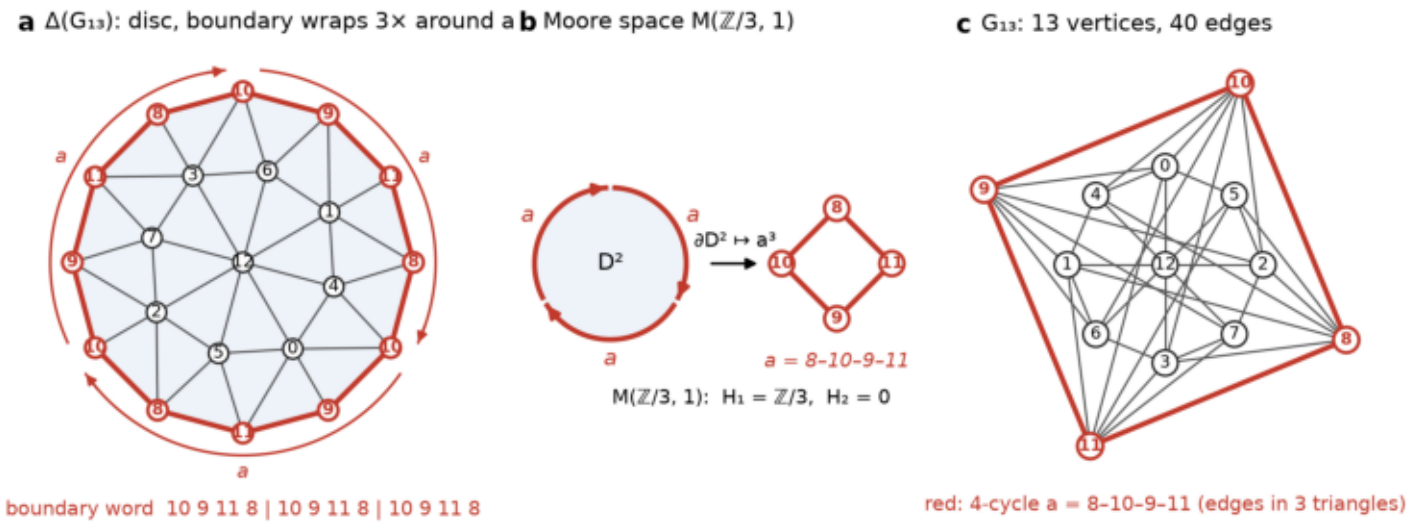


Figure 1. The smallest flag complex with odd torsion: G_{13} = `L?qa``{wttym~o`.
(a) The clique complex $\Delta(G_{13})$ (13 vertices, 40 edges, 28 triangles, K_4-free) cut along the four edges of the 4-cycle 8–10–9–11 (the only edges in 3 triangles) unfolds into a triangulated disc with 21 vertices whose 12-gon boundary reads $(10\ 9\ 11\ 8)^3$: the boundary runs three times, in the same direction, around the cycle *a* = 8–10–9–11 (arcs). (b) Hence $\Delta(G_{13})$ is the Moore space $M(\mathbb{Z}/3, 1)$: a disc whose boundary wraps three times around a circle; $H_1(\Delta; \mathbb{Z}) = \mathbb{Z}/3$, $H_2 = 0$, so Δ is $\mathbb{Z}_2$-acyclic but not contractible. (c) The graph G_{13} itself, drawn with vertex 12 (degree 8) at the centre, its neighbourhood cycle C_8 around it and the 4-cycle *a* outside (red). G_{13} is the unique stuck graph on 13 vertices (Theorem A); its two I_4-closures `L?qa`p{wttym~o` (41 edges) and `L?qahp{wttym~o` (42 edges) have the same homology.

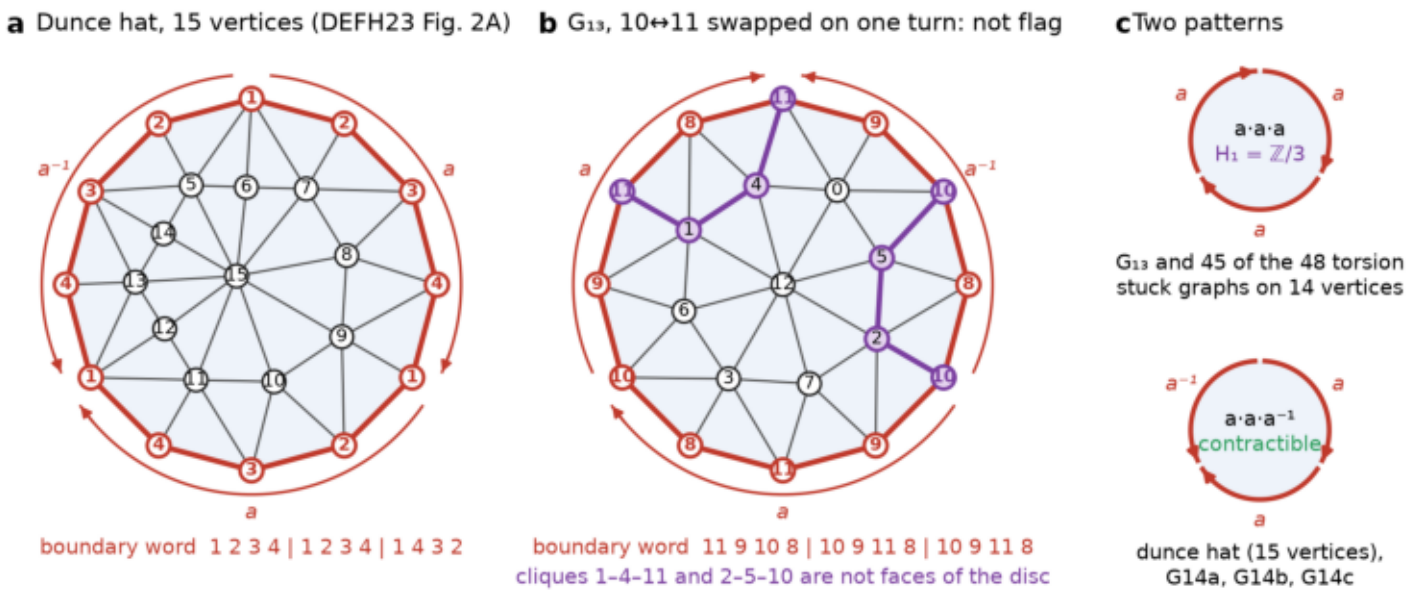


Figure 2. Dunce-hat identifications: the 15-vertex example of DEFH23 versus a 13-vertex attempt.
(a) The dunce hat of DEFH23 Fig. 2A (g6 `N|thHsL`sHoI`RPOwLG`; 15 vertices, 46 edges, 32 triangles) unfolded by the same algorithm: a 23-vertex disc whose 12-gon boundary reads 1 2 3 4 | 1 2 3 4 | 1 4 3 2, i.e. the word *a·a·a⁻¹* on the 4-cycle *a* = 1–2–3–4 (the four edges in 3 triangles). Labels are those of DEFH23 Fig. 2A; the g6 index i corresponds to label (1, 4, 13, 12, 11, 3, 14, 15, 10, 5, 2, 6, 7, 8, 9)[i]. (b) Swapping the labels 10 ↔ 11 on one turn of the boundary of the disc of Fig. 1a gives a disc with boundary word 11 9 10 8 | 10 9 11 8 | 10 9 11 8 = *a⁻¹·a·a*, the dunce-hat pattern, and a contractible 2-complex; but its 1-skeleton (g6 `L?qa``{wtlzM~o`, isomorphic to the canonical form `LtDzQNKmpKXT`V`; 13 vertices, 40 edges) is not flag: the cliques 1–4–11 and 2–5–10 (purple; the two copies of 11, resp. 10, are distinct corners of the disc) are not faces of the disc, and the clique complex contains the two K_4's {1,4,8,11} and {2,5,8,10}. The resulting graph is strong I-contractible, even by vertex deletions alone ([🄰]_VS; verified with `atascadas.c --modo verificar`): there is no flag dunce hat on 13 vertices (Proposition in §6.1). (c) The two identification patterns that occur among all stuck graphs found: a disc glued along *a·a·a* (three turns in the same direction) has $H_1 = \mathbb{Z}/3$ — G₁₃ and 45 of the 48 torsion stuck graphs on 14 vertices (the other three do not unfold to a disc, Fig. 7) — whereas *a·a·a⁻¹* (one turn reversed) gives a contractible complex: the dunce hat and G14a, G14b, G14c.

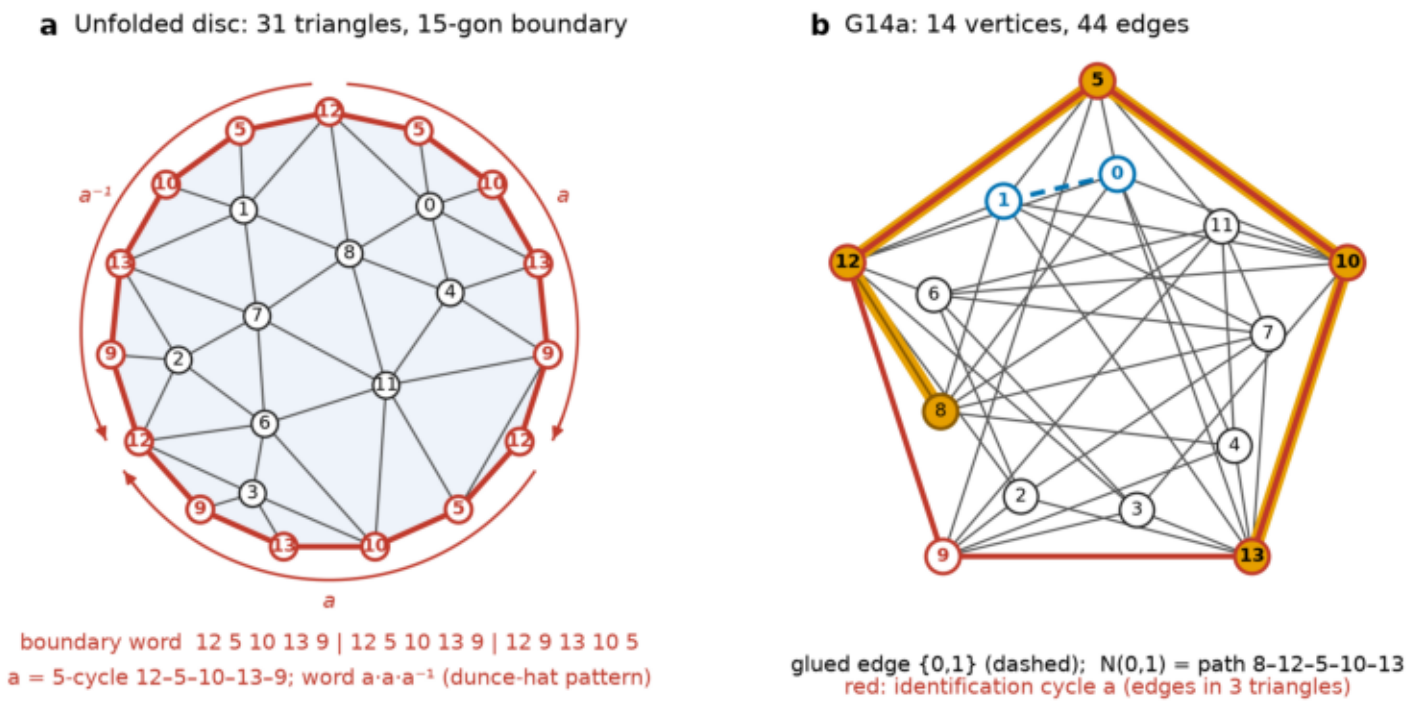


Figure 3. G14a = `M?b@bFHNEk@~|k}U?` (14 vertices, 44 edges, 31 triangles, trivial automorphism group), the first example of 🄰 \ 🄰_S. (a) Unfolded disc (24 vertices); the 15-gon boundary reads 12 5 10 13 9 | 12 5 10 13 9 | 12 9 13 10 5 = *a·a·a⁻¹* on the 5-cycle *a* = 12–5–10–13–9 (the five edges in 3 triangles): the same identification pattern as the dunce hat, hence Δ(G14a) is contractible ($H_1 = H_2 = 0$, Smith normal form). (b) G14a with the cycle *a* as the outer pentagon (red). The non-edge {0,1} (blue, dashed) has common neighbourhood N(0,1) = {5, 8, 10, 12, 13}, the induced path 8–12–5–10–13 (amber), which is strong I-contractible; gluing {0,1} is therefore an I_4 move and G14a + {0,1} = `M_b@bFHNEk@~|k}U?` is strong I-contractible (Fig. 6). G14a itself has no I_S-contractible vertex or edge (stuck).

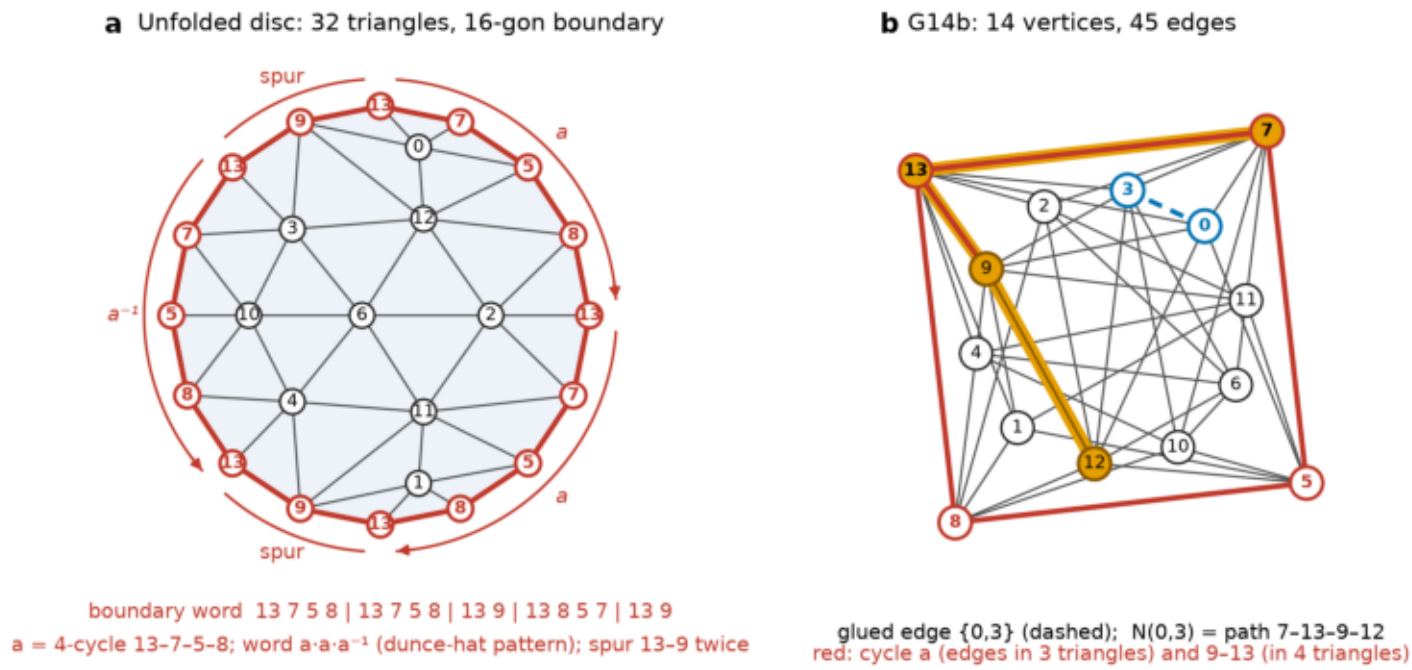


Figure 4. G14b = `M?B@thku?~Lylk}[?` (14 vertices, 45 edges, 32 triangles, |Aut| = 2).
(a) Unfolded disc (25 vertices); the 16-gon boundary reads 13 7 5 8 | 13 7 5 8 | 13 9 | 13 8 5 7 | 13 9: the word *a·a·a⁻¹* on the 4-cycle *a* = 13–7–5–8 interleaved with two traversals (out and back) of the spur 13–9. The edge 13–9 lies in 4 triangles and the four edges of *a* in 3 triangles; all other edges lie in exactly 2 triangles. (b) G14b with *a* as the outer square (red; the spur 9–13 also red). The non-edge {0,3} (blue, dashed) has N(0,3) = {7, 9, 12, 13}, the induced path 7–13–9–12 (amber); G14b + {0,3} = `MCB@thku?~Lylk}[?` is strong I-contractible (Fig. 6b).

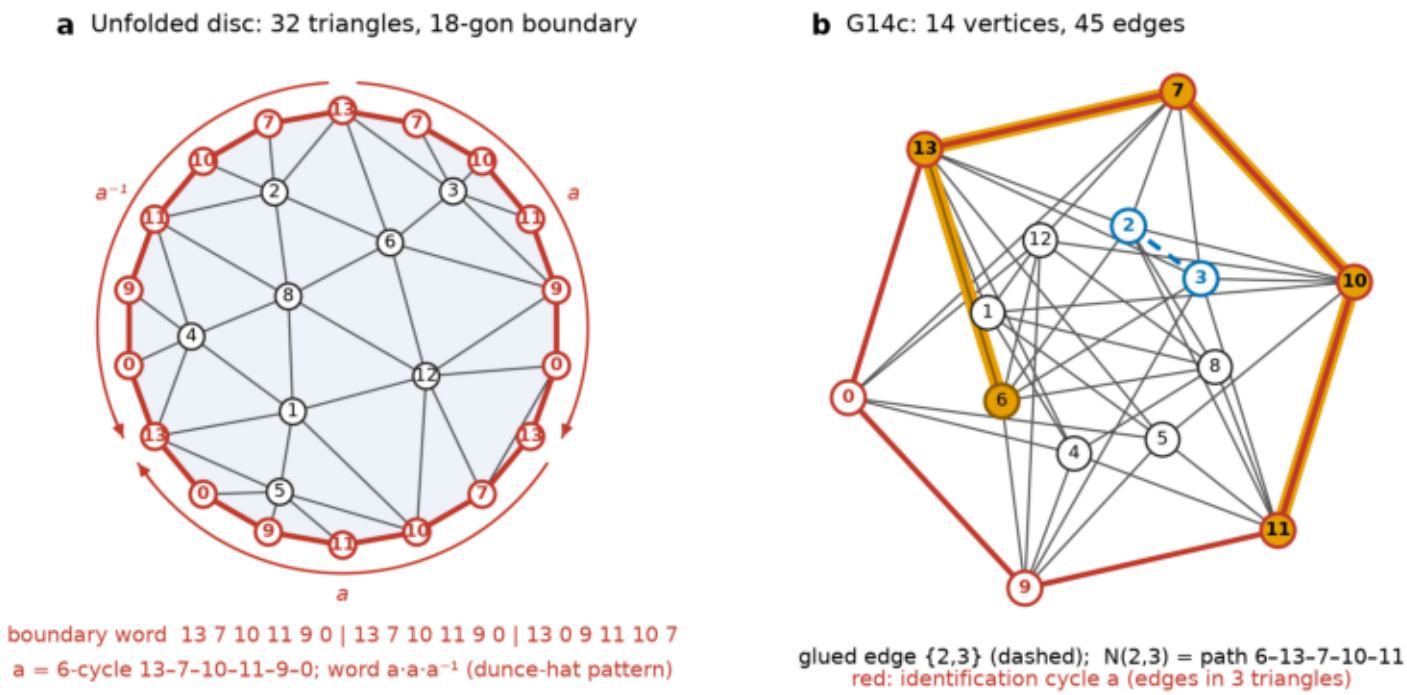


Figure 5. G14c = `M?r@d`ifbiFfo}~o?` (14 vertices, 45 edges, 32 triangles, |Aut| = 2).
(a) Unfolded disc (26 vertices); the 18-gon boundary reads 13 7 10 11 9 0 | 13 7 10 11 9 0 | 13 0 9 11 10 7 = *a·a·a⁻¹* on the 6-cycle *a* = 13–7–10–11–9–0 (the six edges in 3 triangles). (b) G14c with *a* as the outer hexagon (red). The non-edge {2,3} (blue, dashed) has N(2,3) = {6, 7, 10, 11, 13}, the induced path 6–13–7–10–11 (amber); G14c + {2,3} = `M@r@d`ifbiFfo}~o?` is strong I-contractible (Fig. 6c).

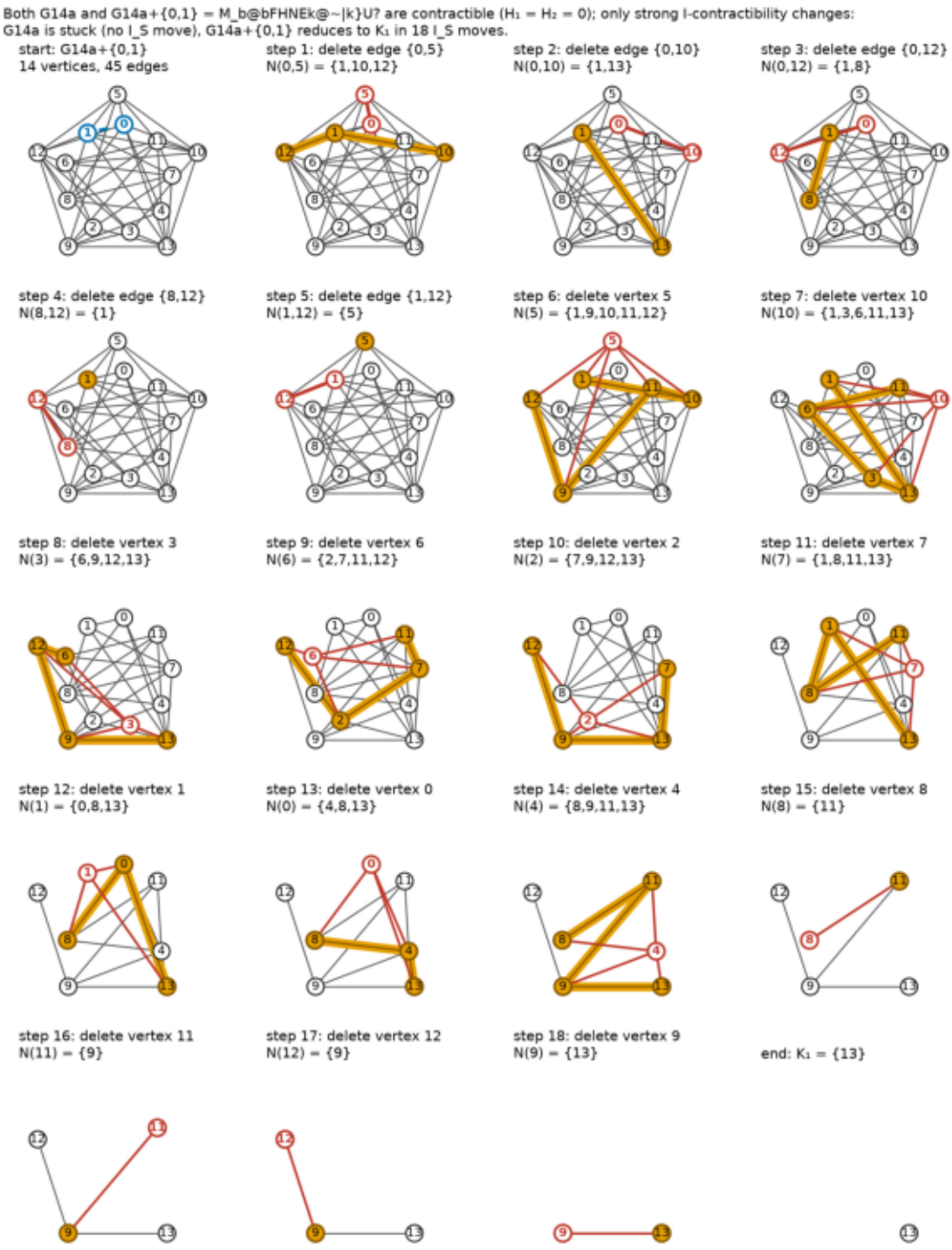


Figure 6. Strong I-contractibility certificate of G14a + {0,1} = `M_b@bFHNEk@~|k}U?` (Theorem D). Starting from G14a with the I_4 edge {0,1} glued (blue, dashed), the 18 deletions listed in `traza_G14_mas01.txt` reduce the graph to a single vertex: steps 1–5 delete edges {u,v} whose common neighbourhood N(u,v) is a cone or K_1 (I_2 moves), steps 6–18 delete vertices whose neighbourhood N(v) is 0-dismantlable (I_1 moves). The header states what does and does not change: both graphs are contractible, only strong I-contractibility differs. In each panel the deleted element is red and its (common) neighbourhood amber (vertices filled, induced edges as a halo); the vertex positions are those of Fig. 3b. Every step was re-verified when drawing (the neighbourhood printed in the title is recomputed from the current graph). The same graph without the edge {0,1} is not strong I-contractible, so {0,1} is an I_S-contractible edge whose removal leaves [A]_S — the minimal instance of the phenomenon of DEFH23 Conjecture 15.



Figure 7. All 51 stuck K_4-free graphs on 14 vertices (exhaustive search `geng -k -C -d4 -D12 14`, 6.49×10^{13} graphs, 2,000,000/2,000,000 slices). Green box: the three contractible ones, G14a, G14b, G14c ($H_1 = H_2 = 0$; the only $\mathbb{A} \setminus \mathbb{A}_S$ stuck graphs), labelled with the length of their identification cycle (5, 4 + spur, 6). Below: the 48 with $H_1(\Delta; \mathbb{Z}) = \mathbb{Z}/3$ (25 with 43 edges, 18 with 44, 4 with 45, 1 with 47), sorted by edge count; under each drawing its g6 string. Red edges lie in ≠ 2 triangles; for 45 of the 48 torsion graphs they form the identification cycle *a* (a 4-, 5- or 6-cycle, drawn as the outer polygon) and the boundary word is $*a^3*$; two graphs (*) unfold into an annulus and one (**) has cut edges forming two disjoint 4-cycles. The three contractible graphs are the only ones with pattern $*a{\cdot}a{\cdot}a^{-1}*$.

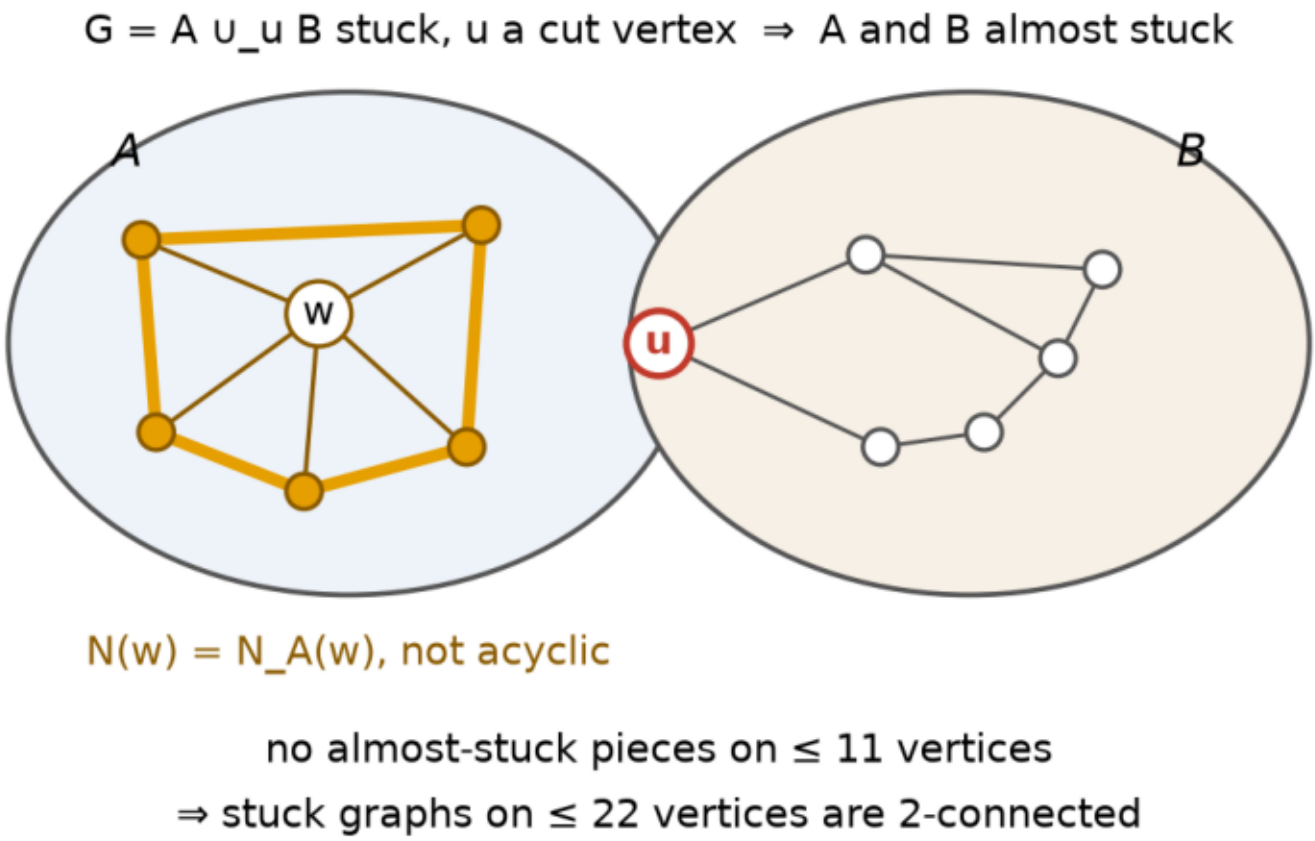
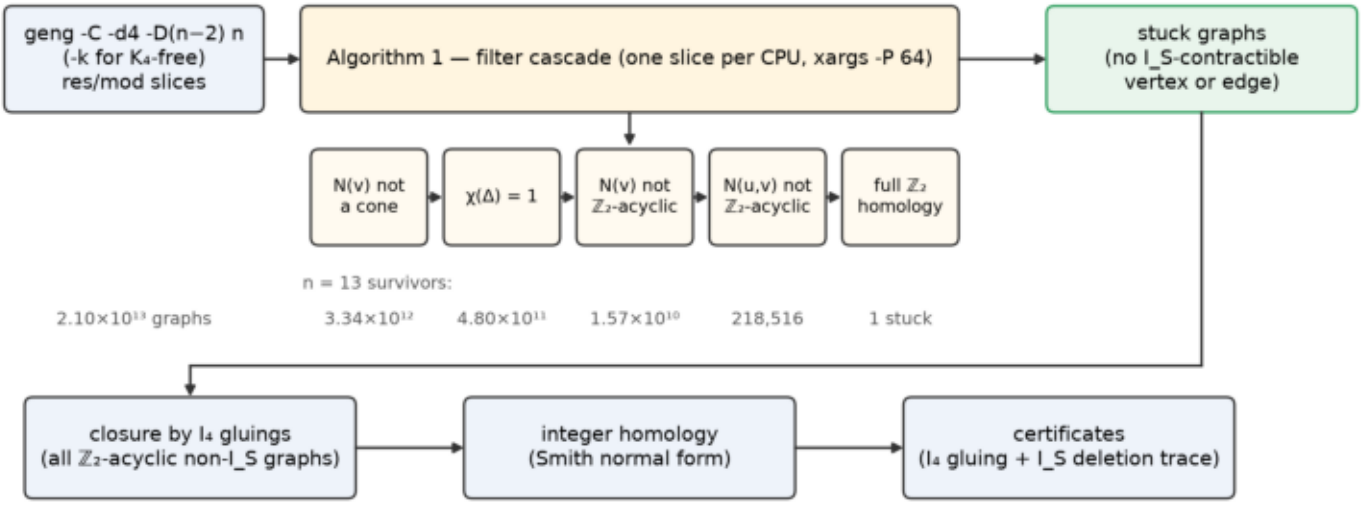


Figure 8. The 2-connectivity lemma (Lemma 3.2).
If a stuck graph G has a cut vertex u, write G = A ∪_u B. For every w ∈ A − u the whole neighbourhood N(w) lies in A, so N(w) = N_A(w) is not acyclic, and every edge of A keeps its common neighbourhood: A and B are *almost stuck* ($\mathbb{Z}_2$-acyclic, every vertex other than u with non-acyclic neighbourhood, every edge stuck). The computation finds no almost-stuck piece on ≤ 11 vertices, hence every stuck graph on ≤ 22 vertices is 2-connected; consequently it has no bridge, every edge lies in ≥ 2 triangles, the minimum degree is ≥ 4 and the maximum degree is ≤ n − 2, which justifies the generator options `-C -d4 -D(n−2)`.



YUCA (ACARUS, UNISON; SLURM, 64-core AMD EPYC nodes):

n = 12	1.64×10¹¹ graphs	8 min	448 CPU	row 12 of the table (ℤ₂-acyclic = I_S)
n = 13	2.10×10¹³ graphs	7 h 10 min	448 CPU	200,000 slices → 1 stuck graph ($H_1 = \mathbb{Z}/3$)
n = 14, K₄-free	6.49×10¹³ graphs	≈ 33 h	896 CPU, 2 accounts	2,000,000 slices → 51 stuck graphs

Figure 9. The distributed computation on YUCA (ACARUS, Universidad de Sonora).
Graphs are generated by `geng` in res/mod slices and each slice is processed by one CPU running Algorithm 1, a cascade of increasingly expensive tests (no neighbourhood is a cone → χ(Δ) = 1 → no N(v) ℤ₂-acyclic → no N(u,v) ℤ₂-acyclic → full ℤ₂-homology of Δ); survivors are the stuck graphs. For n = 13 the cascade reduces 2.10 × 10¹³ connected graphs with 4 ≤ deg ≤ 11 to 3.34 × 10¹², 4.80 × 10¹¹, 1.57 × 10¹⁰, 218,516 and finally 1 stuck graph. Post-processing: closure under I₄ gluings (all ℤ₂-acyclic non-🄸_S graphs of that order), integer homology by Smith normal form, and certificates (I₄ gluing plus I_S deletion trace). Runs: n = 12, 1.64 × 10¹¹ graphs, 8 min on 448 CPUs (row 12 of the table); n = 13, 2.10 × 10¹³ graphs, 7 h 10 min on 448 CPUs, 200,000 slices; n = 14 K₄-free, 6.49 × 10¹³ graphs, ≈ 33 h on 896 CPUs under two SLURM accounts, 2,000,000 slices.